

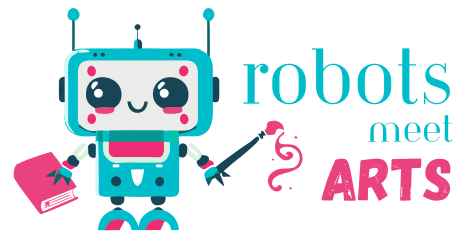


Co-funded by
the European Union

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FR01-KA220-SCH-
000151881

MASTERPIECE MEMORY



**Subject:**

Art & design

Content:

Looking up artworks and learning to work with a floor robot.

**Computational practices:
choose what fits**

- Decomposing
- Pattern recognition
- Abstraction
- **Algorithms**
- **Problem solving**

Unplugged/**plugged:**

**Preparation time:**

20 min

**Age:**

10-12 years

**Duration:**

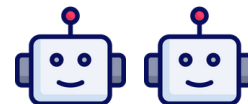
120 min

**Location:**

Large classroom with sufficient floor space

**Group distribution:**

4 groups of 5 pupils 20 pupils in total

**Difficulty****After this lesson, the children will be able to...**

- program a floor robot to complete a simple task.
- break down a task into clear steps to program a floor robot.
- use logical reasoning to determine the correct path for the robot.
- debug and adjust their robot's path when it does not reach the correct card.
- gather and write down key information about an artwork.
- match artworks with their corresponding details.
- collaborate in small groups to complete different tasks.
- develop curiosity and appreciation for art through interactive play.



Materials needed: for 20 kids

station 1 & 2 (identical):

- 2x info sheet 1
- 2x floor robots
- 2 x 6: paint splatters

station 3 & 4 (identical):

- 2 x info sheet 2
- 2 x 20 memory cards with works of art
- 2 x 20 blank memory cards
- 4 x tablets or Chromebooks or laptops for researching artworks

Memory game with robots:

- 2 x floor robot
- 2 x memory cards:
 - cards with artwork
 - cards with info of the pupils

attached to this lesson:

- info sheets 1 and 2 (copy 2 x)
- 20 images of works of art with correction (copy 2 x)
- 20 blank memory cards (copy 2 x)

Helpful links:

With Google Lens, you can identify artworks:

- <https://lens.google/>

How to code Cubetto:

- <https://www.youtube.com/watch?v=iuUDkpvlwcU>

How to code Beebot:

- https://www.youtube.com/watch?v=iNg5QLqYP_Y

Preparation and teacher notes

Cut out the cards and set up the stations before the lesson begins.

You can also customize the artwork cards to fit a different theme, such as a museum you recently visited with your class, and include those artworks as part of the follow-up activity.

Timing & Rotation

- Each station lasts 20 minutes.
- After the first round, students from Stations 1 & 2 switch places with those in Stations 3 & 4.

Choose an educational robot.

There are several robots you can use for this activity. If your class or school has a floor robot, you can use it during this activity. The only requirement is that the children must be able to program the robot to move forward, left, right, and or backward.

Some robots you can use for this activity include: Cubetto, Bee-bot, Lego Spike, Photon, and others.

Warm up 20'

Robot Adventure Story: Tell a short, exciting story about a robot lost in an art museum, needing help to navigate to the correct paintings. "Today, you will be programming robots to help them find their way!" We are going to see how we can help this robot out of trouble. Then, the teacher will walk around the stations and explain the goal of each one.

Stations 1 & 2 (identical): Floor Robot Colour Mixing 40'

1. Introduce the floor robot and demonstrate how to program it to move forward, backward, left, and right. More information in the links above.
2. Explain the activity:
 - Six cards are placed randomly on the floor or a large table.
 - Each pair of cards represents a colour mixture (e.g., Card 1: Yellow + Blue, Card 2: Green).
 - Students must program the robot to start at the first card and navigate to the correct colour outcome.
 - If the robot does not stop at the correct card, they must adjust their program and try again.
3. Encourage collaboration and problem-solving as students refine their programs.

Preparation and teacher notes

Memory cards are provided with the correct answer so the teacher does not have to look everything up in advance. The aim is for students to look up this information themselves. If necessary, the students can correct themselves afterwards using the correction cards.

Make sure students write clearly on the memory cards. If necessary, write the name of the artwork in capital letters.

Stations 3 & 4 (identical): Art Recognition Challenge

1. Provide each group with 2x 20 cut-out art cards featuring famous artworks and 2x 20 corresponding answer cards.
2. Students select an artwork card and research the following questions using a computer:
 - What is the name of the artwork?
 - Who is the artist?
 - Where is it exhibited?
3. They write the answers on two separate answer cards, ensuring that each artwork has two identical sets of answer cards.
4. Each student researches one artwork, so by the end, the group has five completed sets.

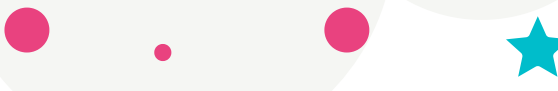
tip: Have students use google lens.

Via Google Images:

- Go to Google Images.
- Upload a photo or search for an image.
- Click the Lens icon (top right of the image).
- Lens will analyse the image and provide results.

Via Google Chrome:

- Open Google Chrome.
- Right-click on an image.
- Choose 'Find image with Google Lens'.
- You will get information about the object or text in the image.



Preparation and teacher notes

Conclusion: Memory Game with Robots

1. Gather all students together and discuss their experiences:
 - What challenges did they face?
 - What did they learn about programming and art?
 2. Introduce the final game:
 - Two memory games are set up, each involving two groups competing against each other.
 - Each group uses the same type of floor robot they used in the first activity.
 - Players take turns programming the robot to move from one card (A) to another (B).
 - They then flip the cards:
 - If the artwork matches the correct answer card, they collect the pair and earn a point.
 - If the match is incorrect, they flip the cards back over.
 3. The game continues until all artwork cards have been correctly matched with their corresponding answer cards.
 4. Celebrate the winning team and review the key lessons learned!
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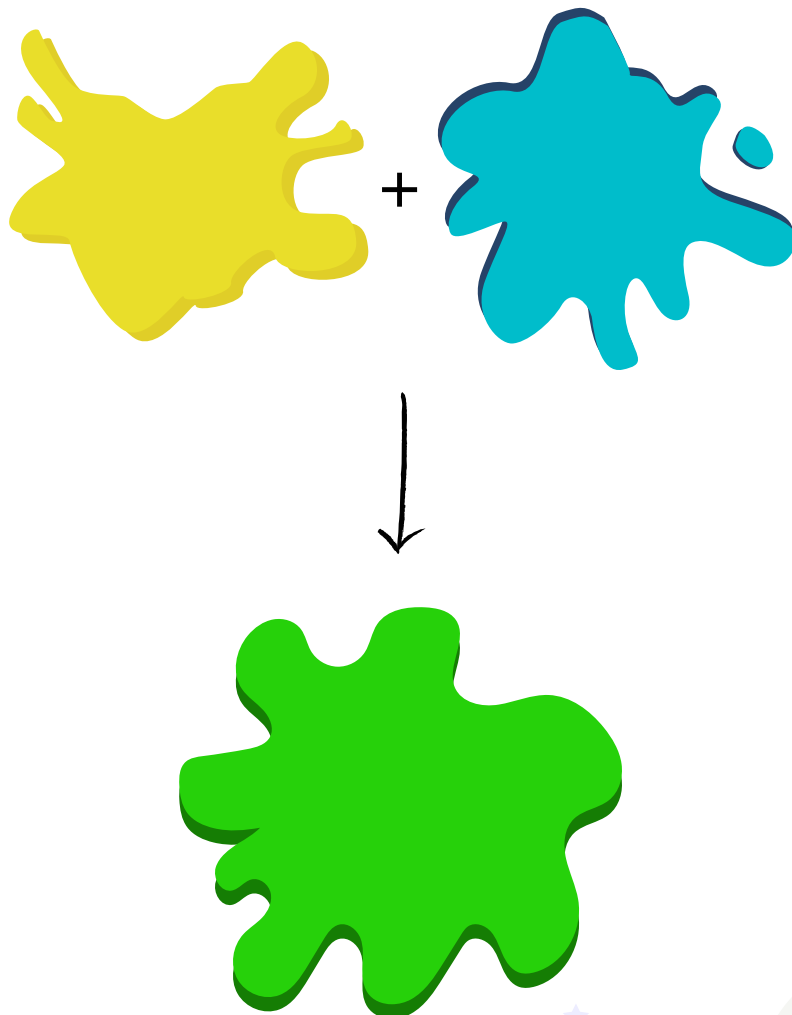
Info sheet 1

Teaching a robot to walk.

Can you let the robot walk to the right colour?

What colour is created when you mix two colours? Together with your classmate, think carefully about the steps you need to take! The six cards are scattered across the table.

Program the robot to start at the first card and navigate to the correct colour outcome. If the robot does not stop at the correct card, adjust your program and try again!



Info sheet 2

Discovering Art in Europe.

Will you help make a big memory of the most beautiful art in Europe?

Look at the artworks on the 20 memory cards and find information about them on the internet:

- name artwork
- artist
- place of art

Write down the answers on an empty memory card. Study them carefully because after this you should be able to link the card with the artwork to the card with the correct information.

Each student finds the information for one card and writes it on two blank memory cards.

Here is an example:



De intrige

James Ensor

**Koninklijk Museum
voor Schone Kunsten
Antwerp**



**Poseidon of
Artemision**

Calamis

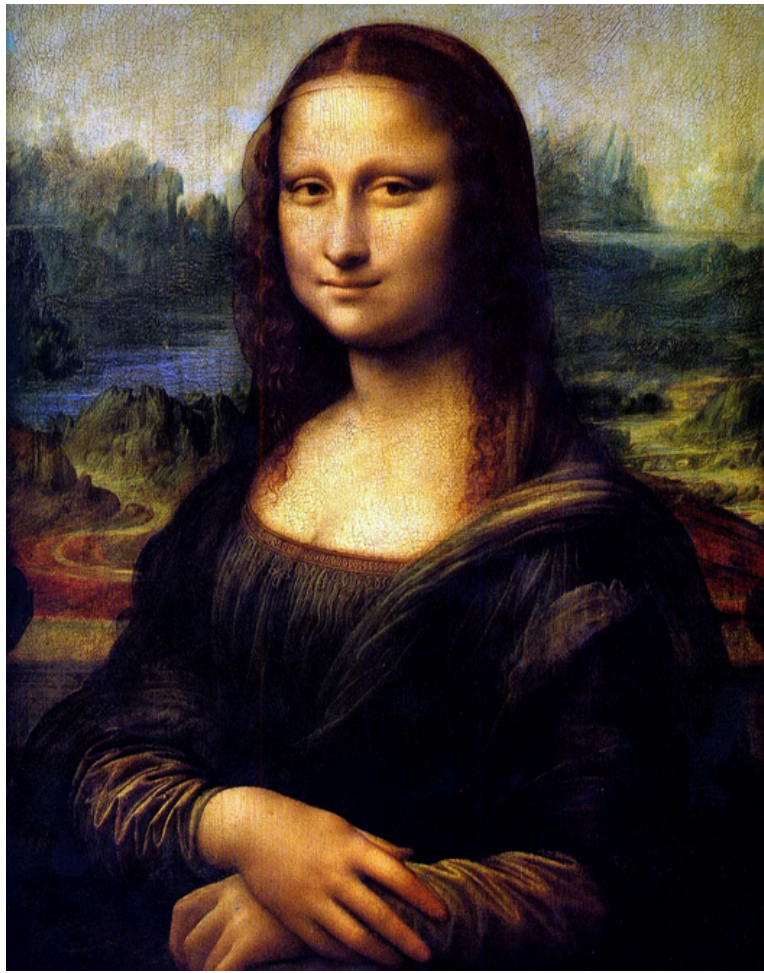
**National
archeological
museum Athens**



Sagrada Familia

Antoni Gaudi

Barcelona



Mona Lisa

Leonardo Da Vinci

Le Louvre Paris



The observatory

Vincent Van Gogh

**Museum of Modern
Art in New York**



Peasant wedding

**Pieter Bruegel de
oude**

**Kunsthistorisches
Museum in Vienna**



Galatea of the Spheres

Salvador dali

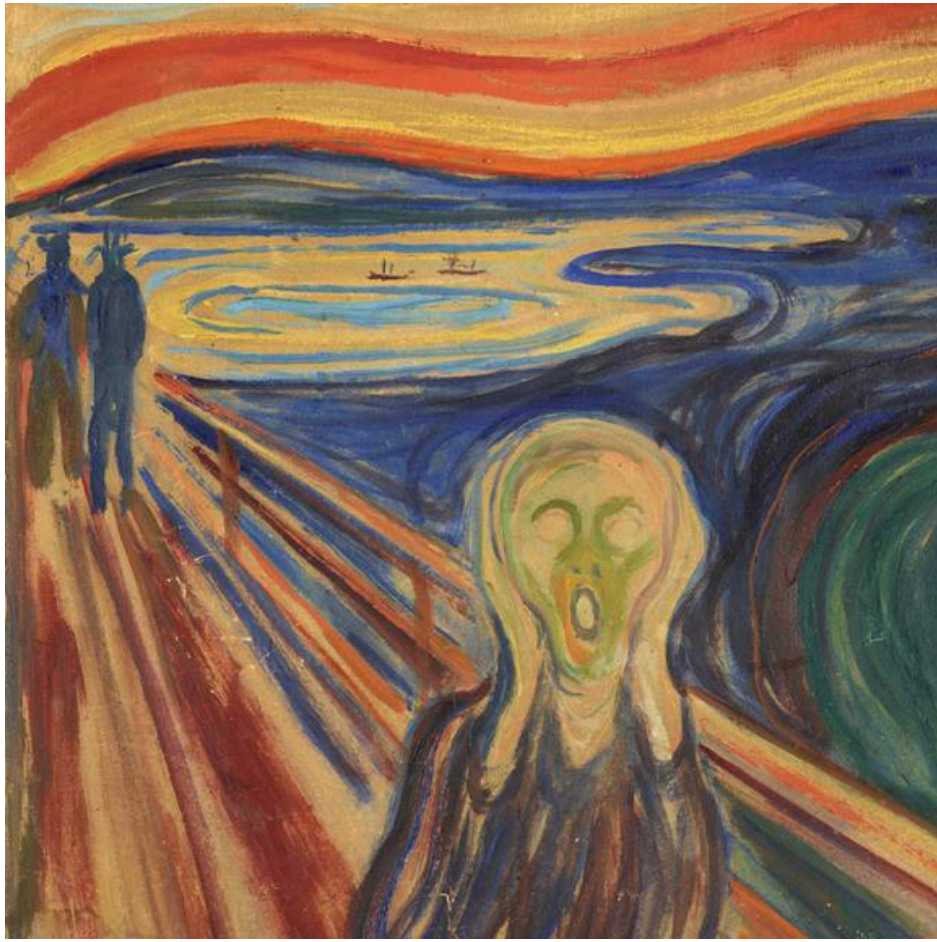
**Dali museum
Figueras**



Eifel Tower

Gustave Eifel

Paris



The Scream

Edvard Munch

**National Museum of
Art, Architecture
and Design, Oslo**



Guernica

Pablo Picasso

**Musea Reina Sofia,
Madrid**



The Little Mermaid

Edward Eriksen

Copenhagen



Atomium

**André and Jean
Polak**

Brussels



David

Michelangelo

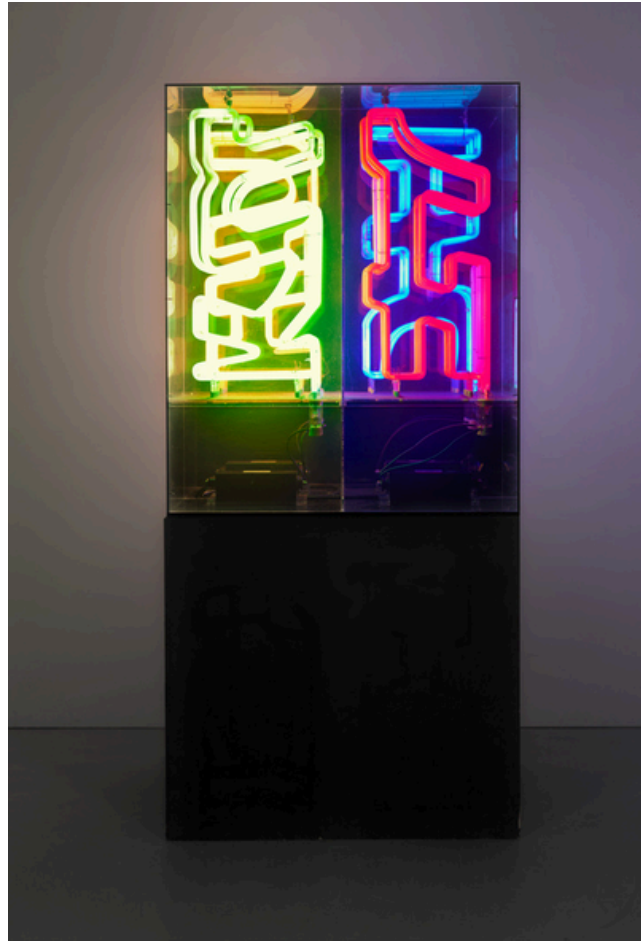
**Galleria
dell'Accademia,
Firenze**



Little Pissing Man

**Hiëronymus
Duquesnoy**

Brussels



Chryssa

**Study for the Gates
#2**

**Whitney Museum of
American Art**



Girl with Balloon

Banksy

London



Water Lilies

Claude Monet

**Musée de
l'Orangerie, Paris**



Iberian Lynx

Bordallo II

Lisbon



Holocaust Memorial

Peter Eisenman

Berlin



Fork of Vevey

**Jean-Pierre Zaugg
and C.Toda**

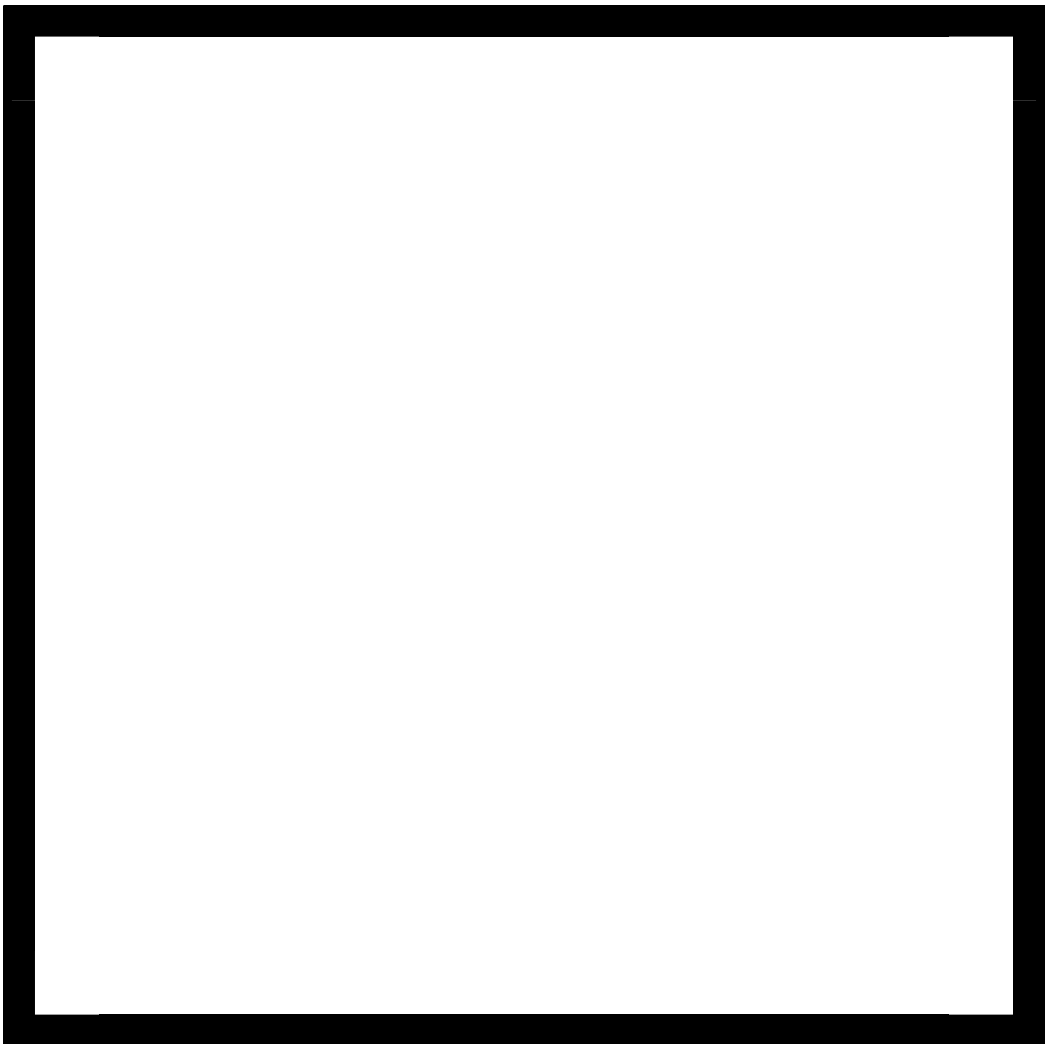
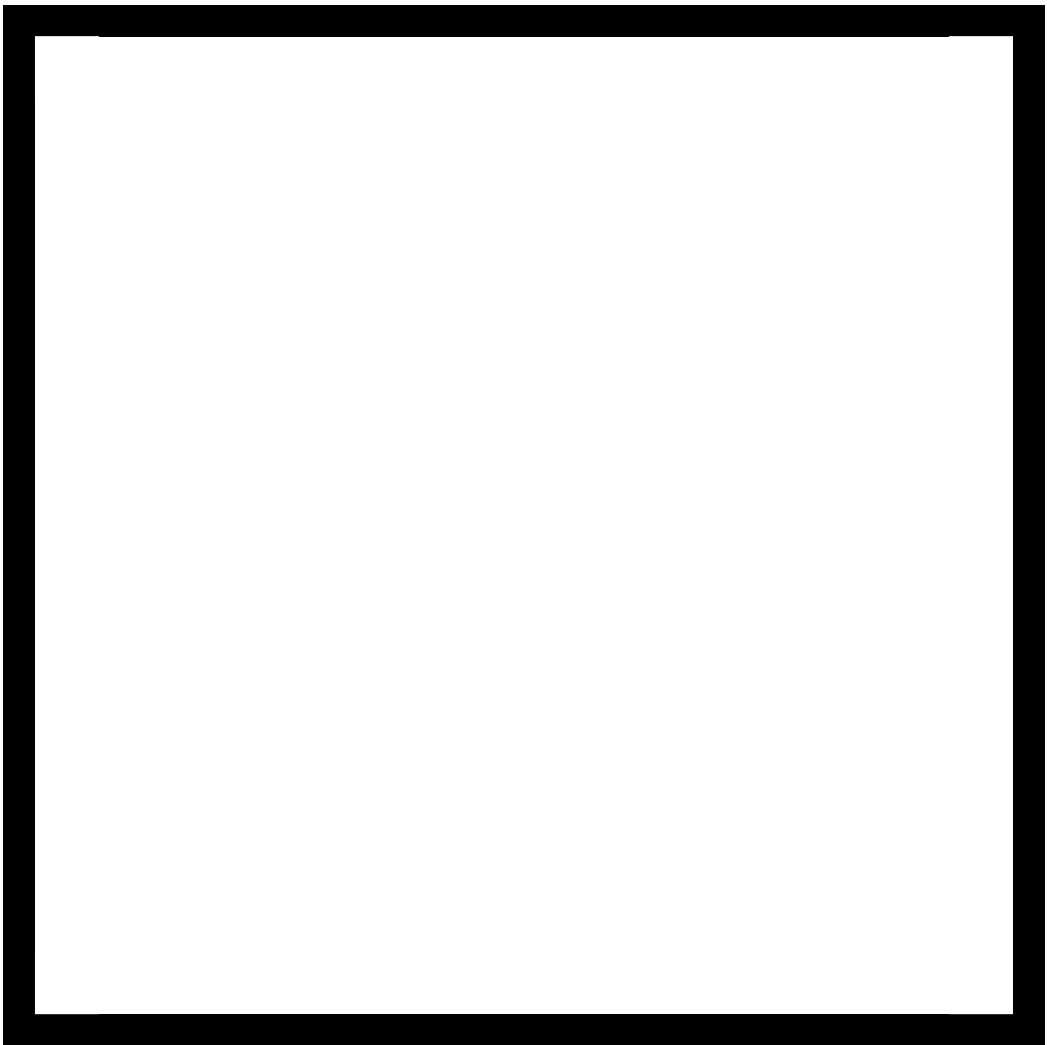
Lake Geneva

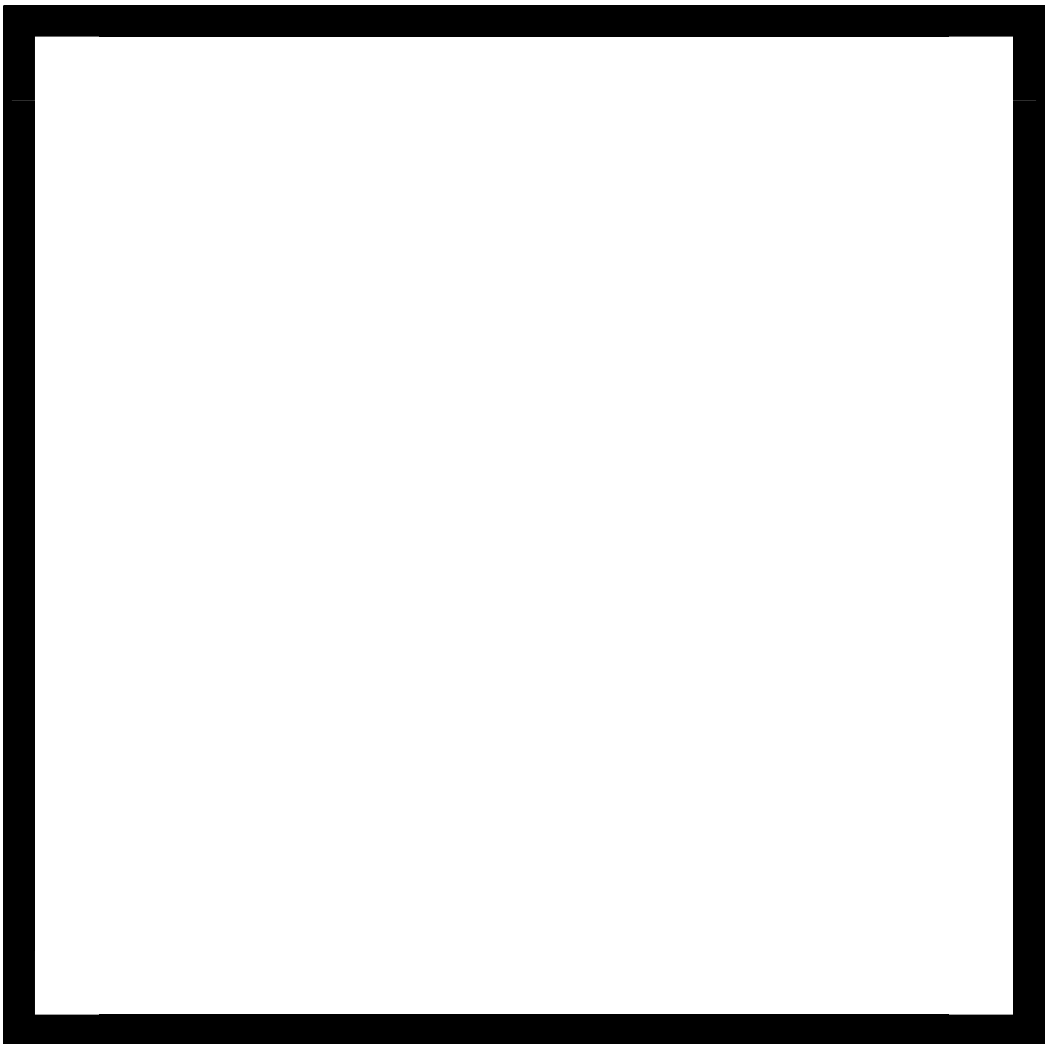
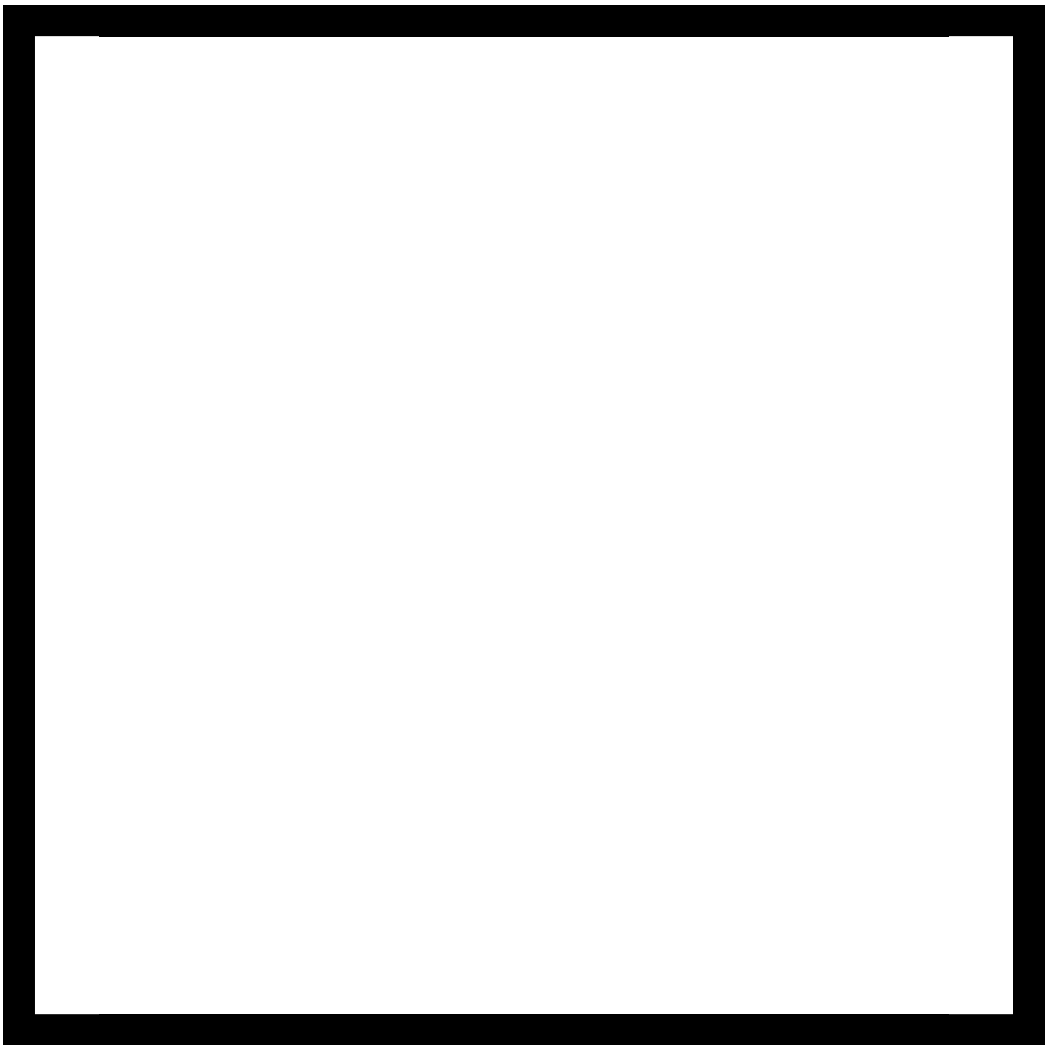


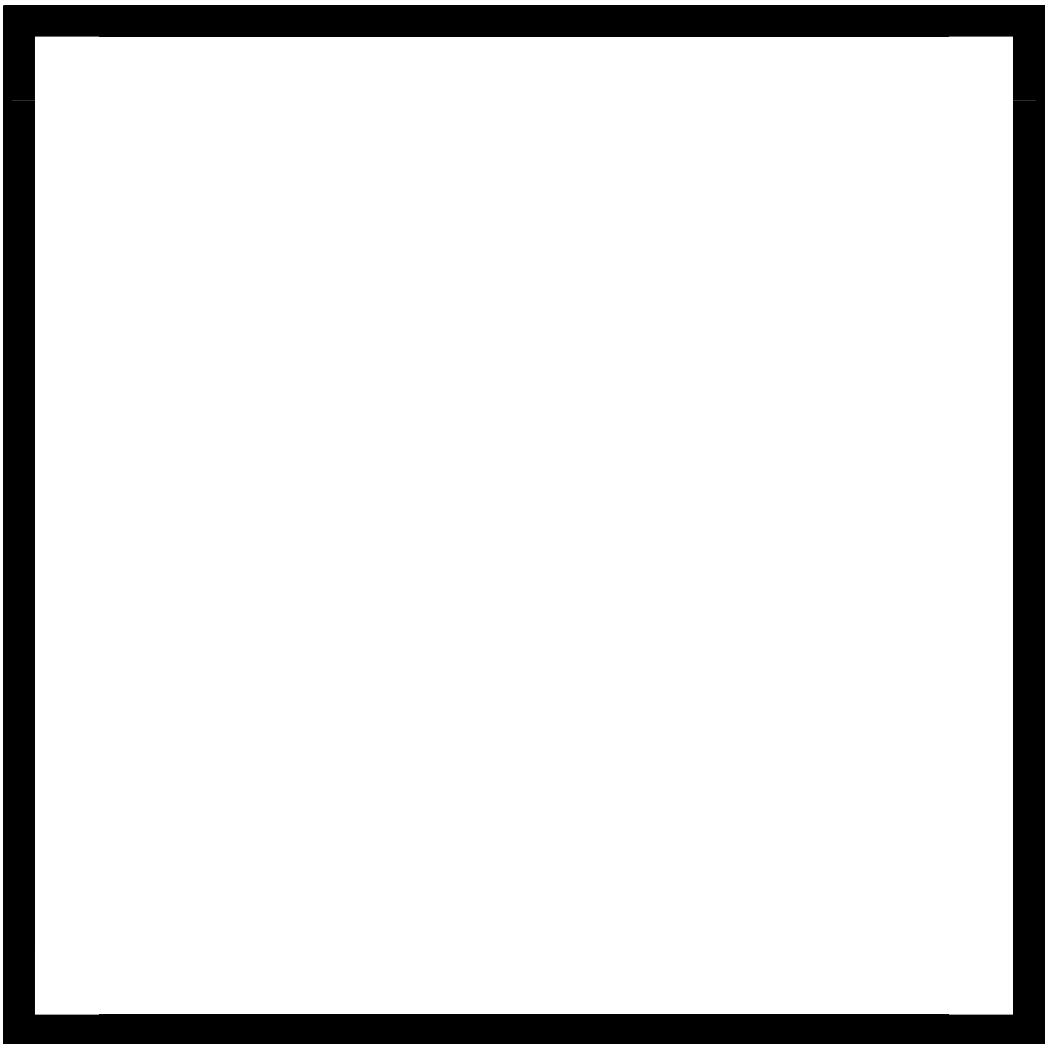
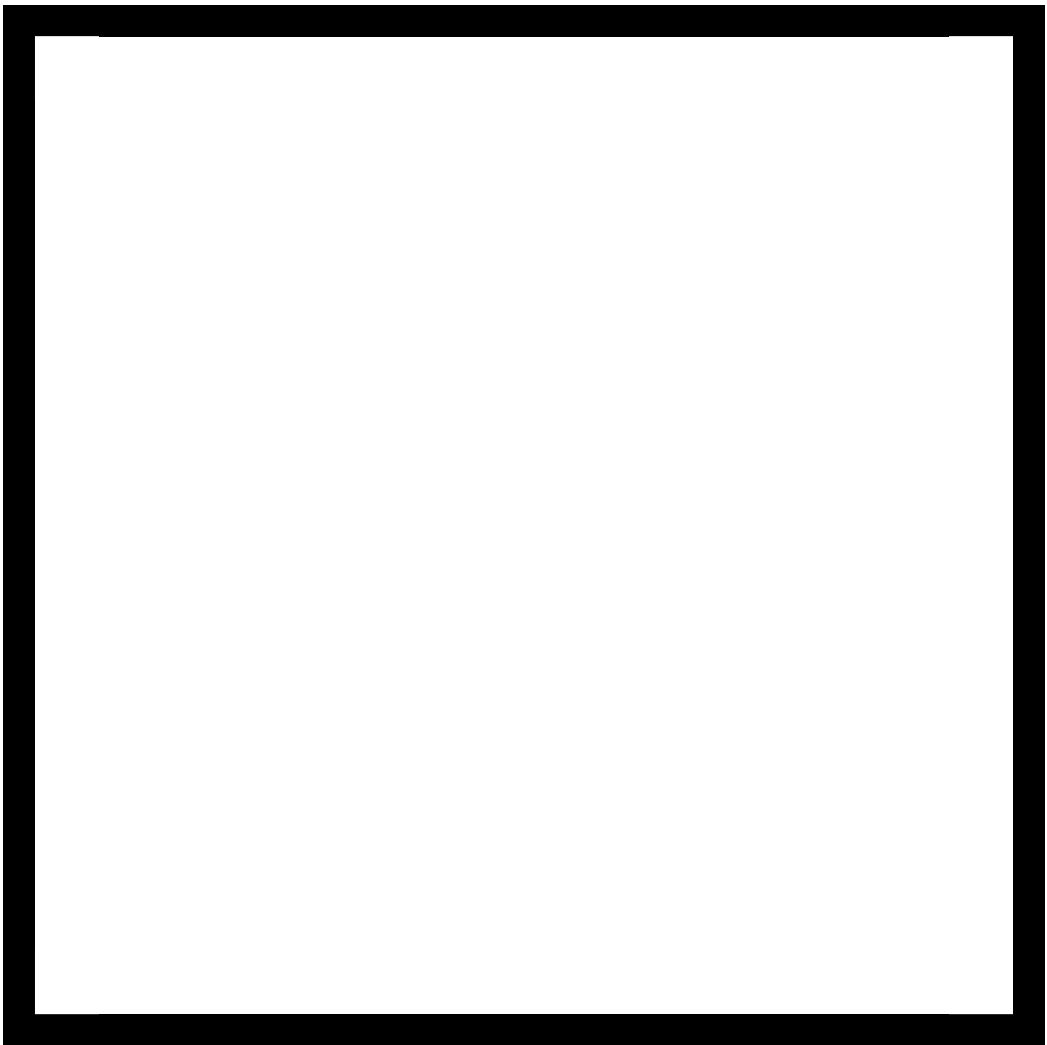
The Kiss

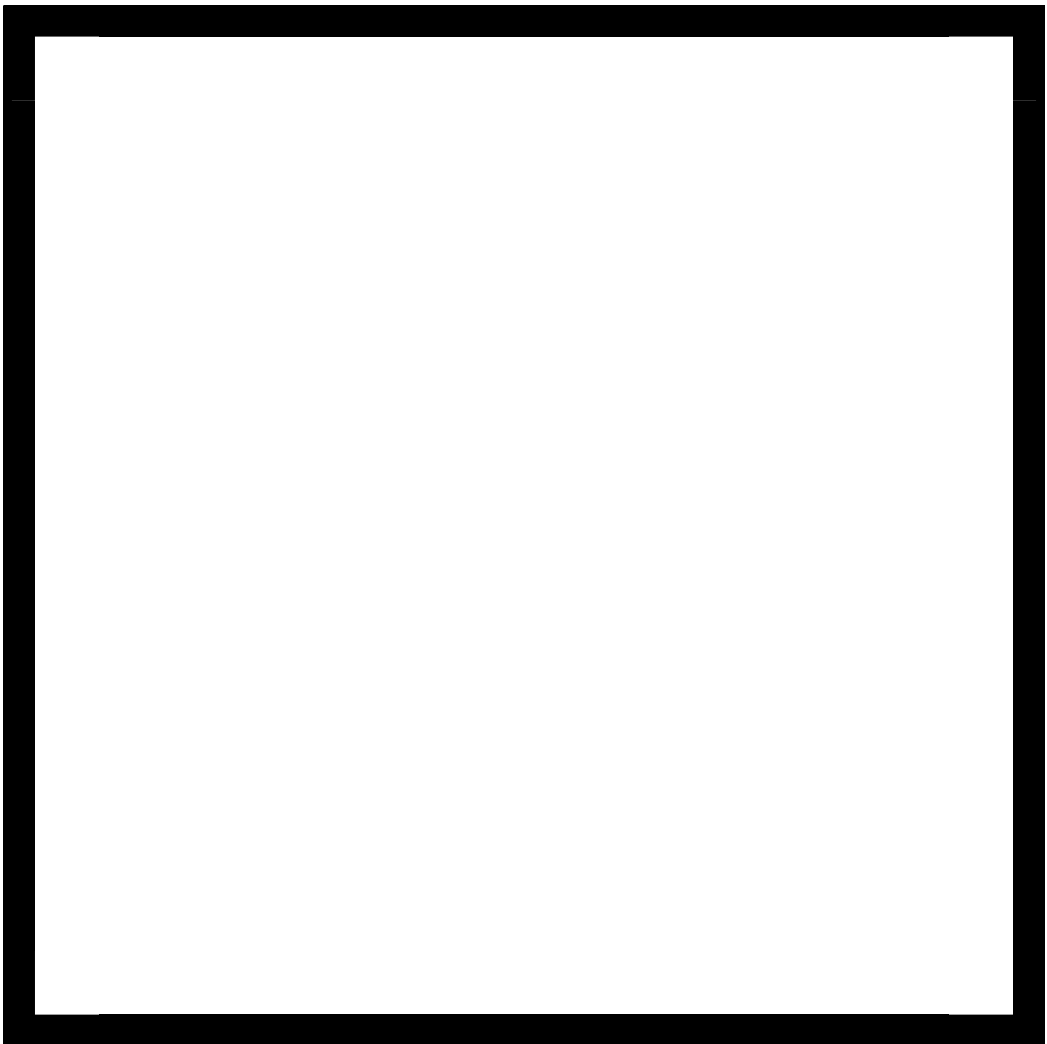
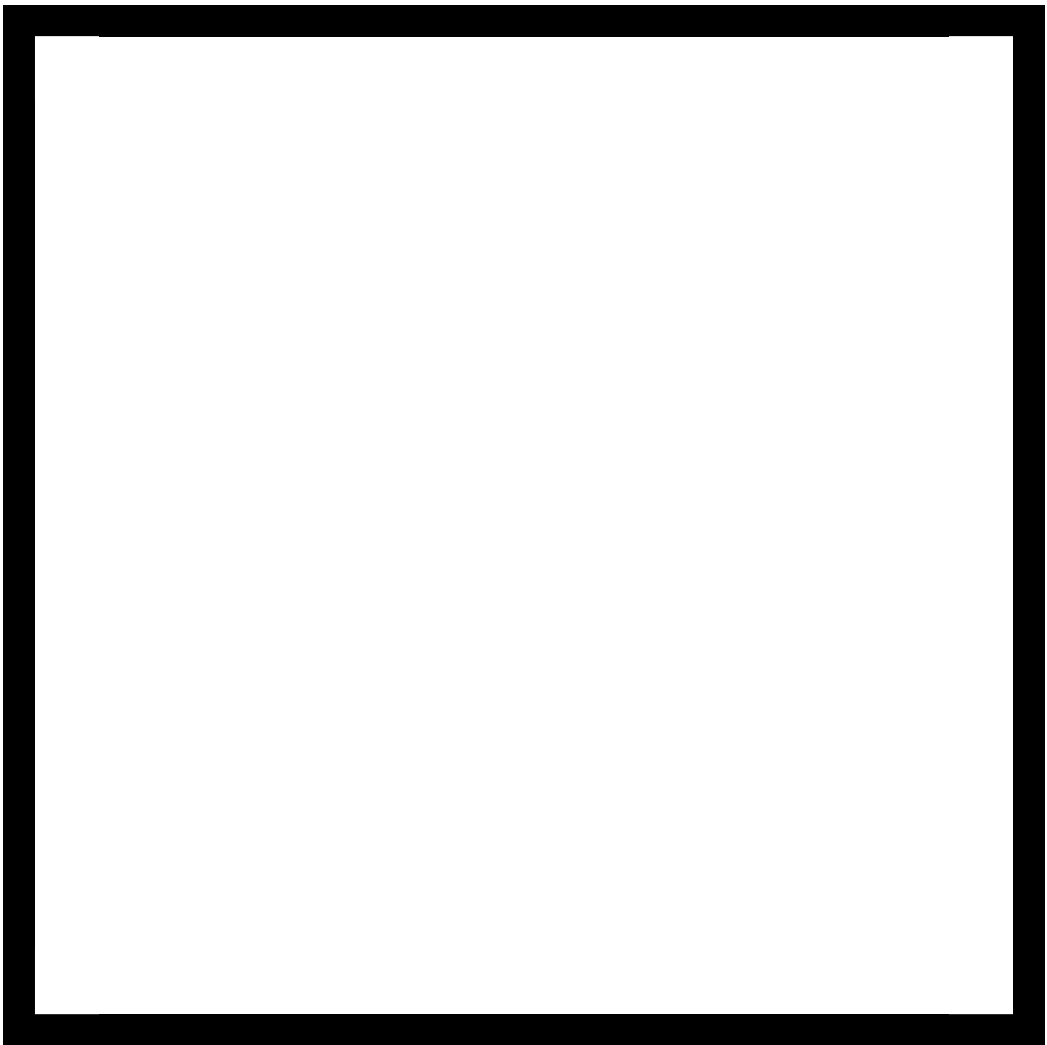
Gustav Klimt

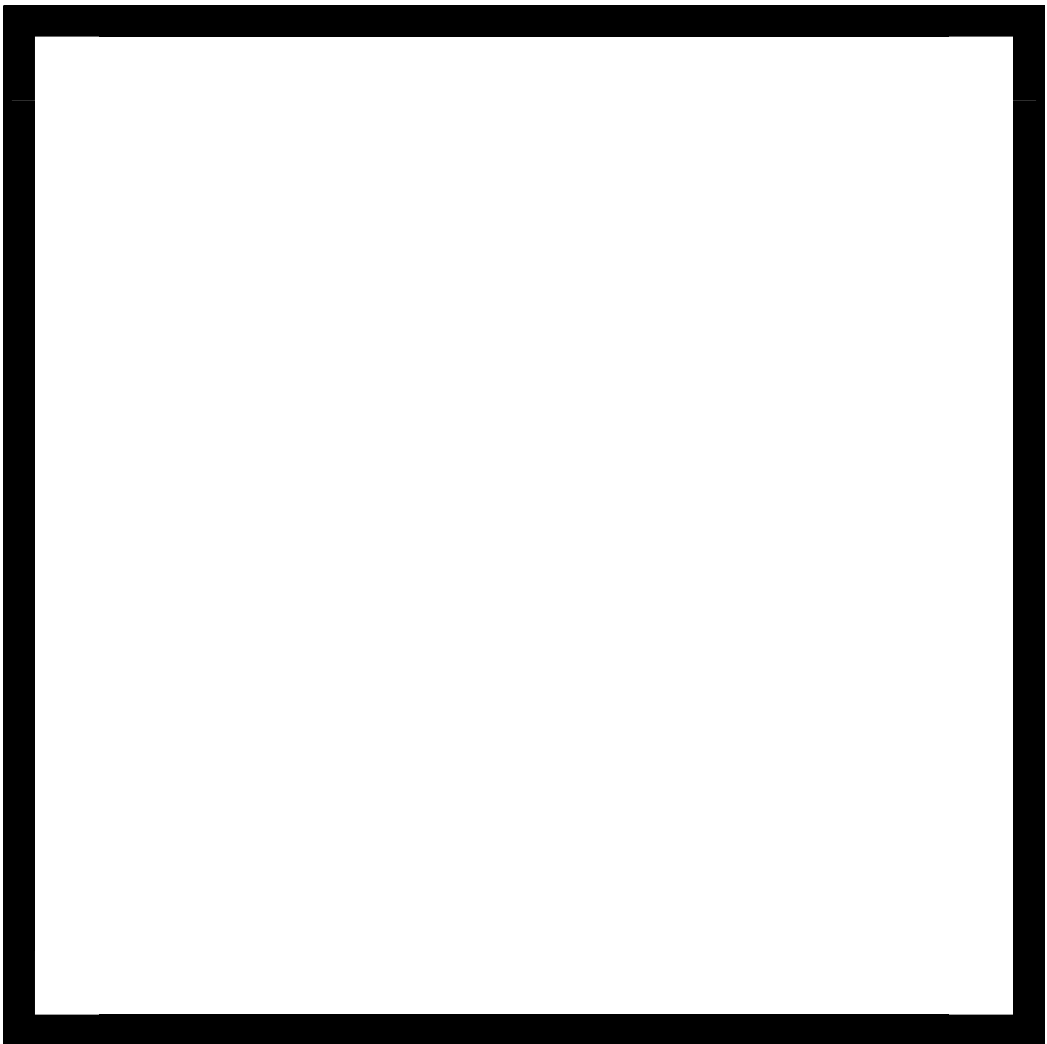
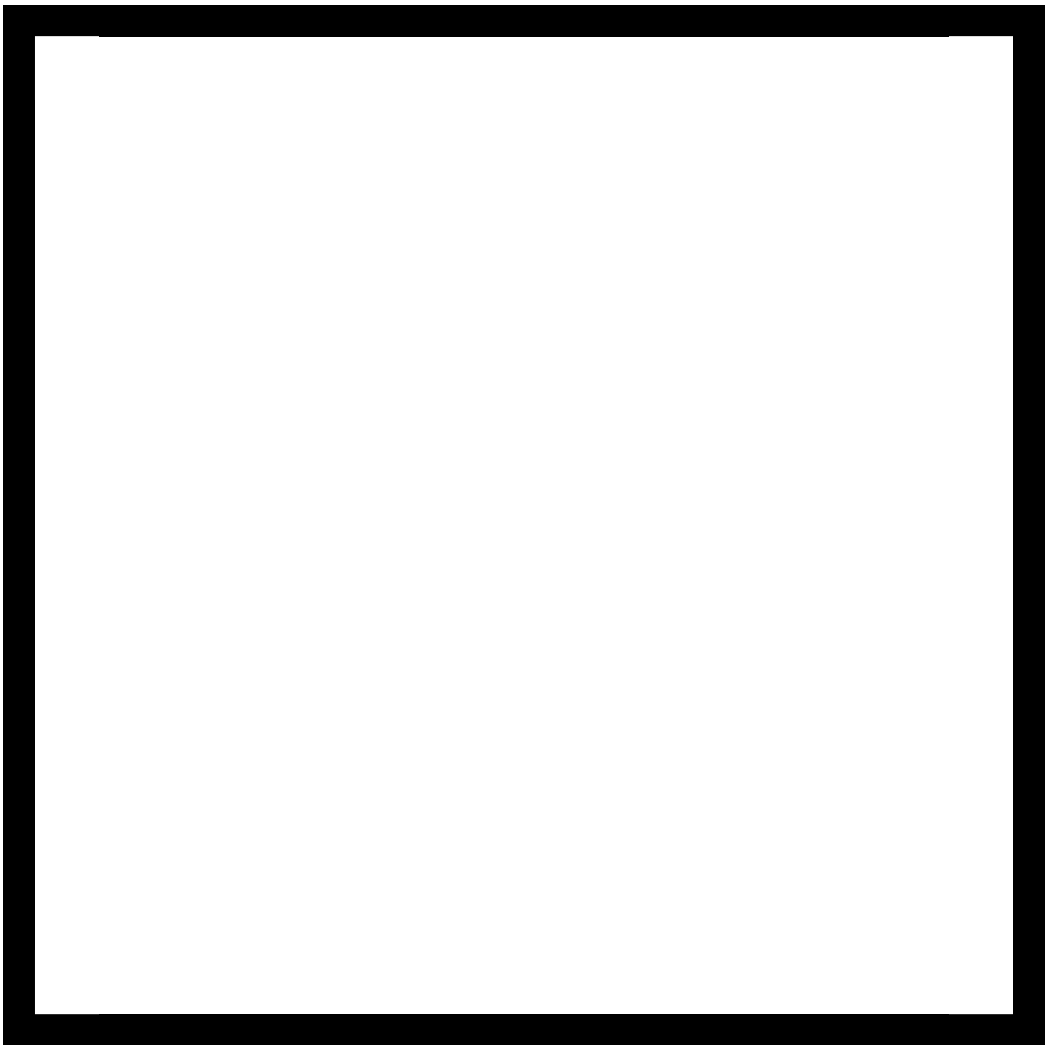
**Österreichische
Galerie Belvedere,
Vienna**

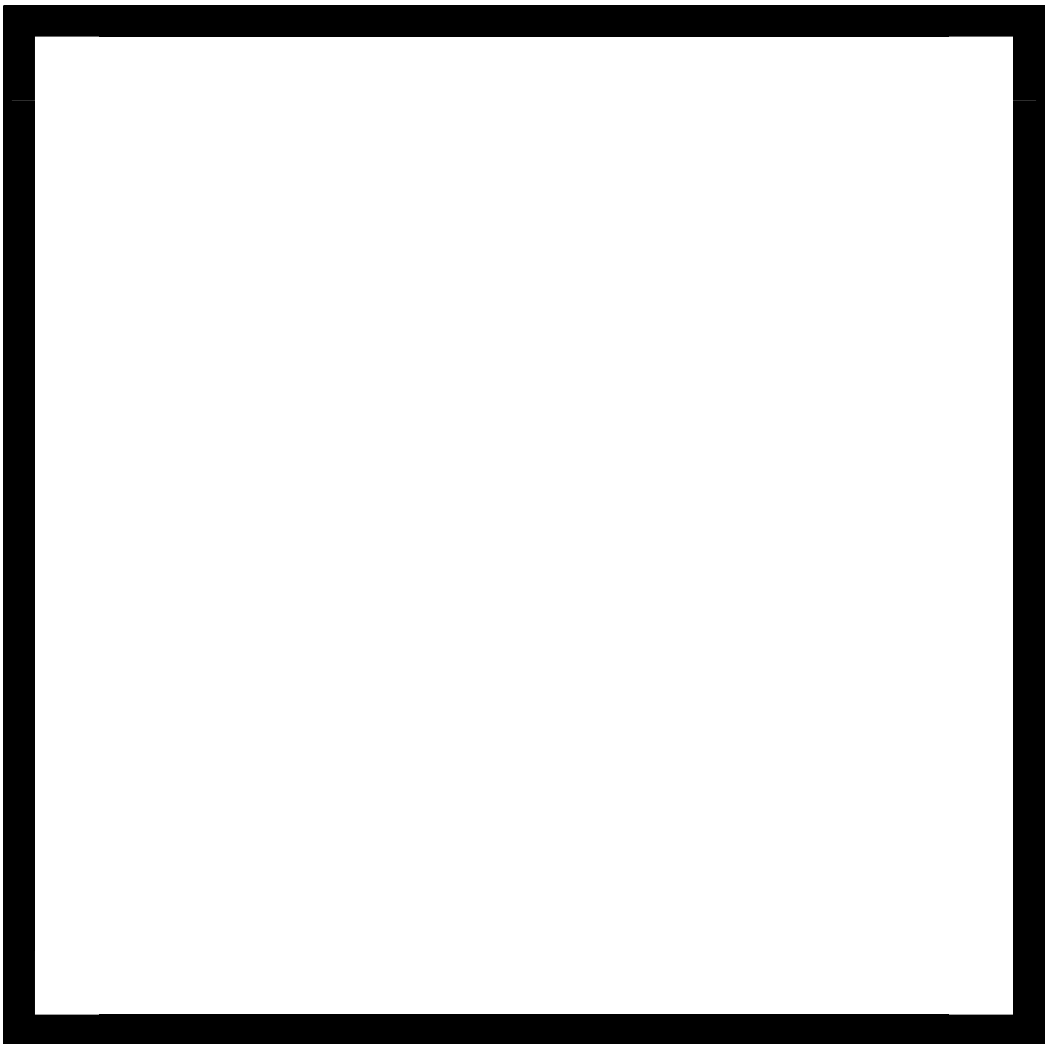
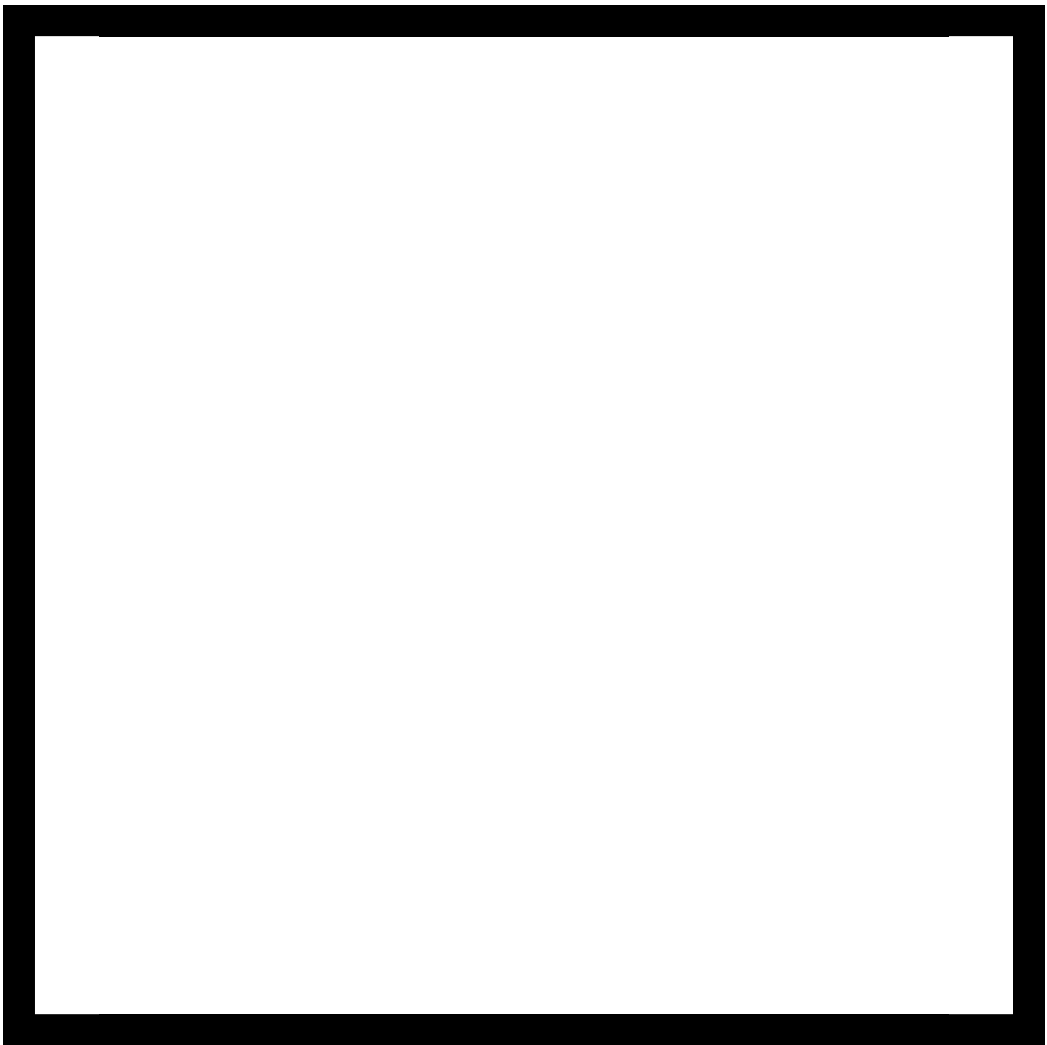


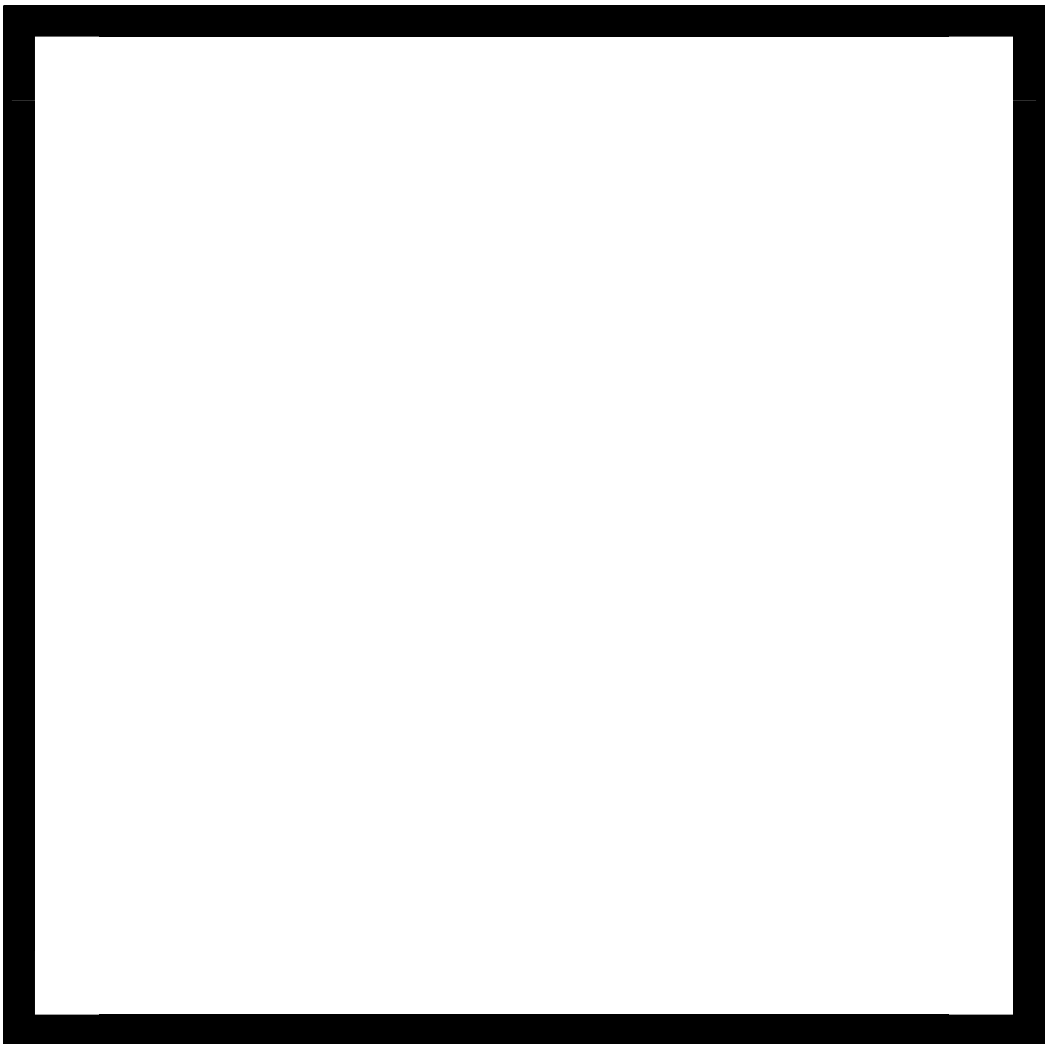
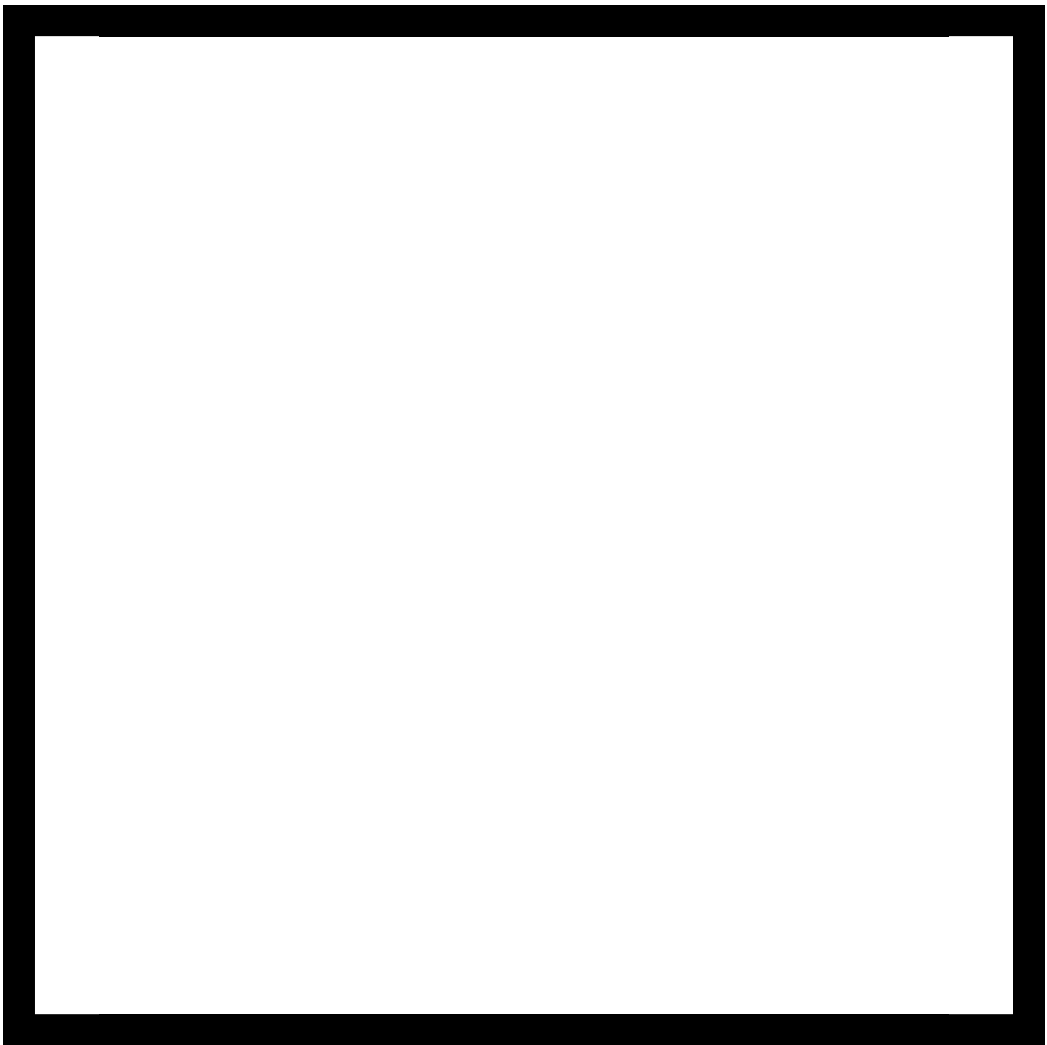


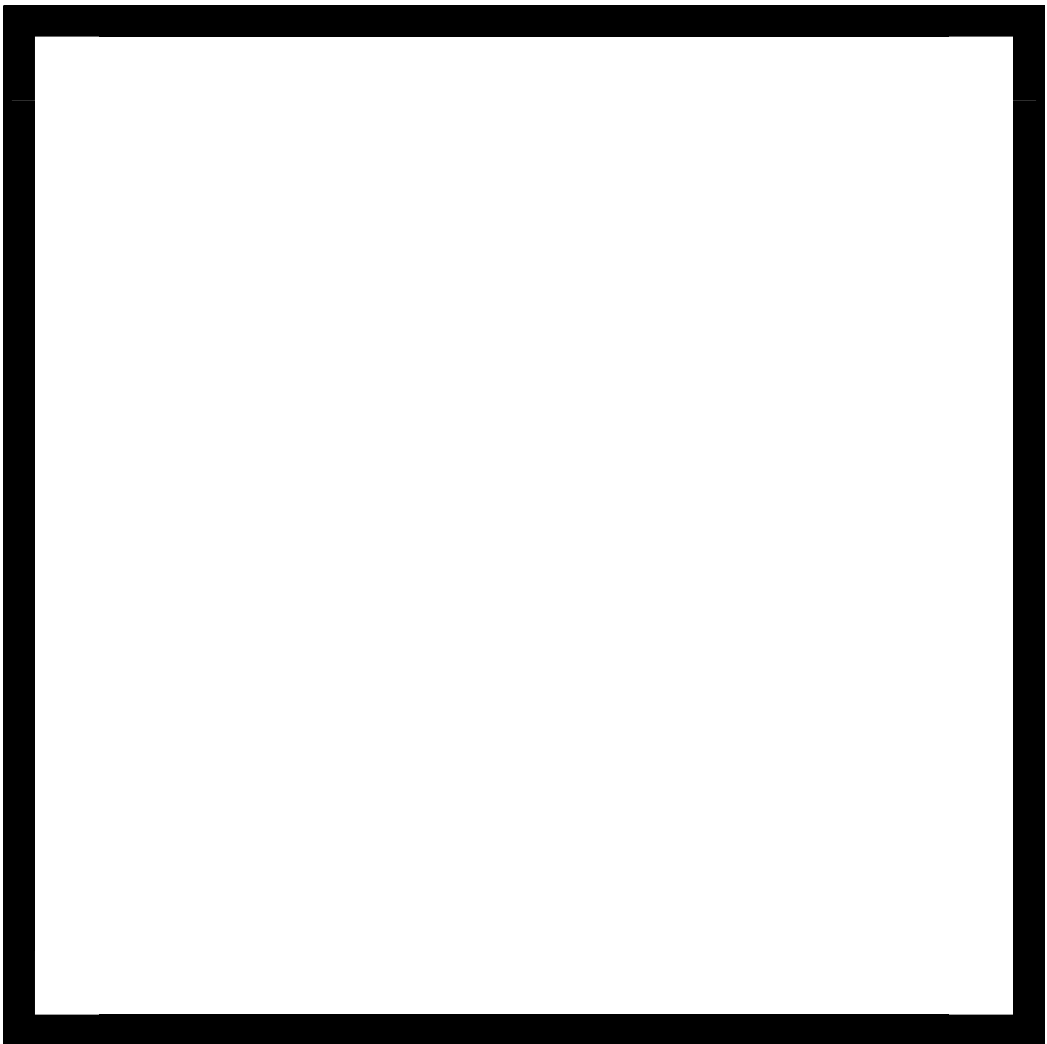
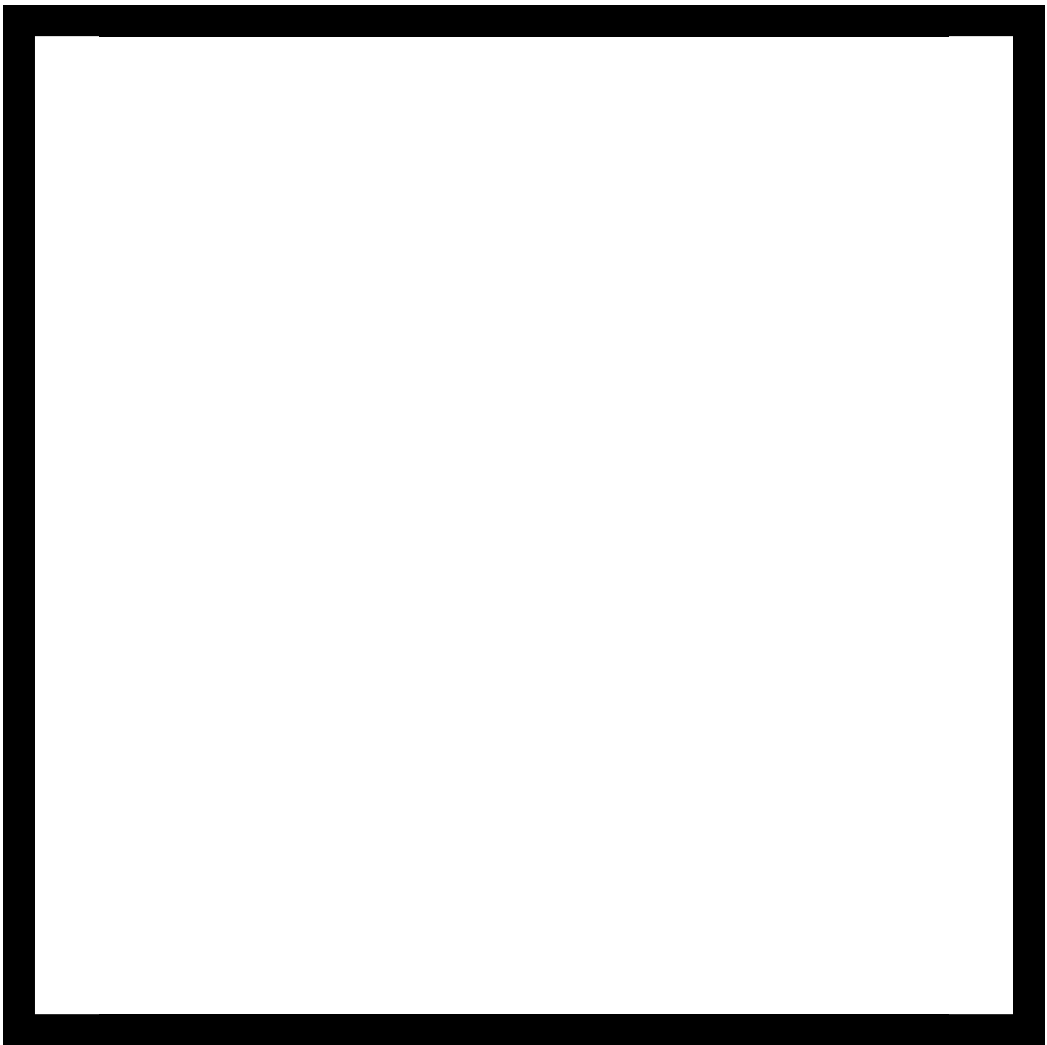


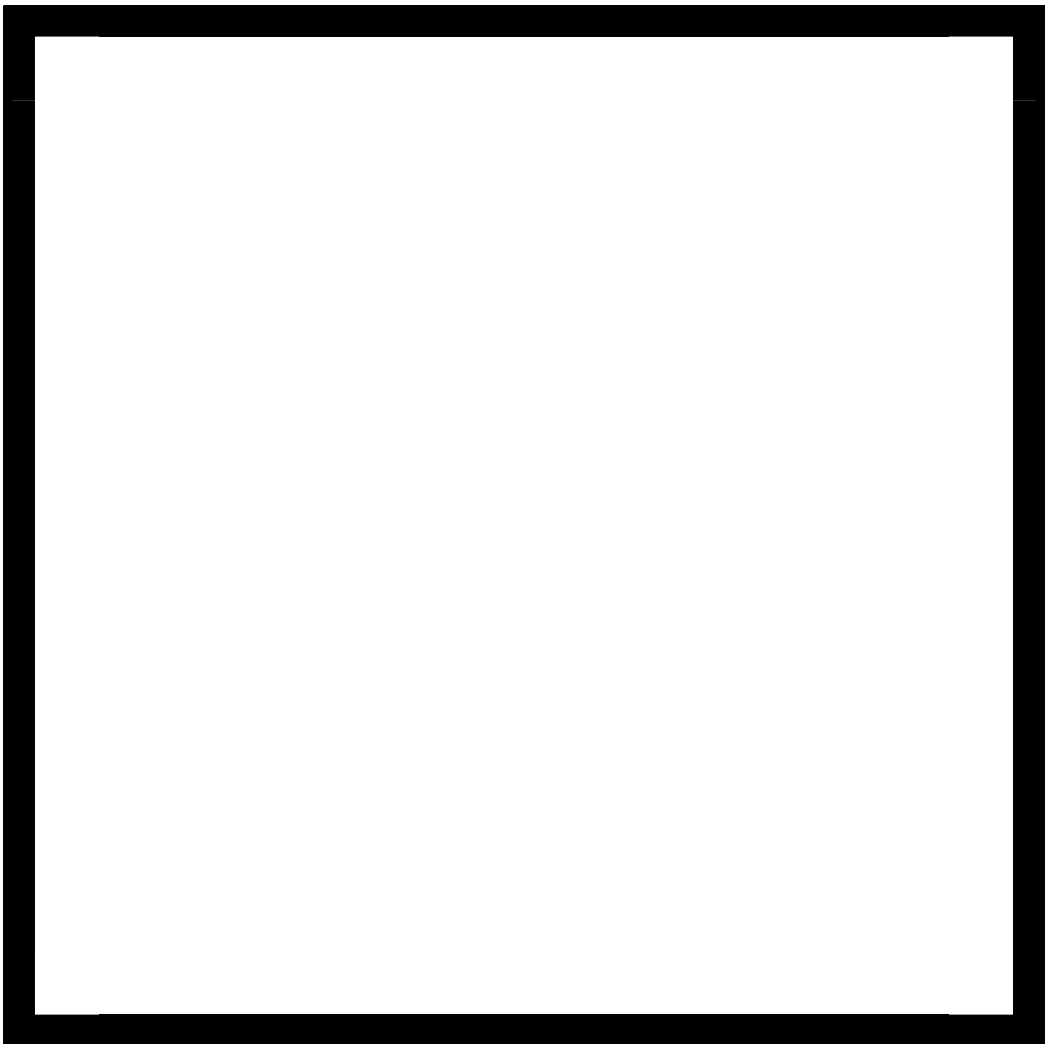
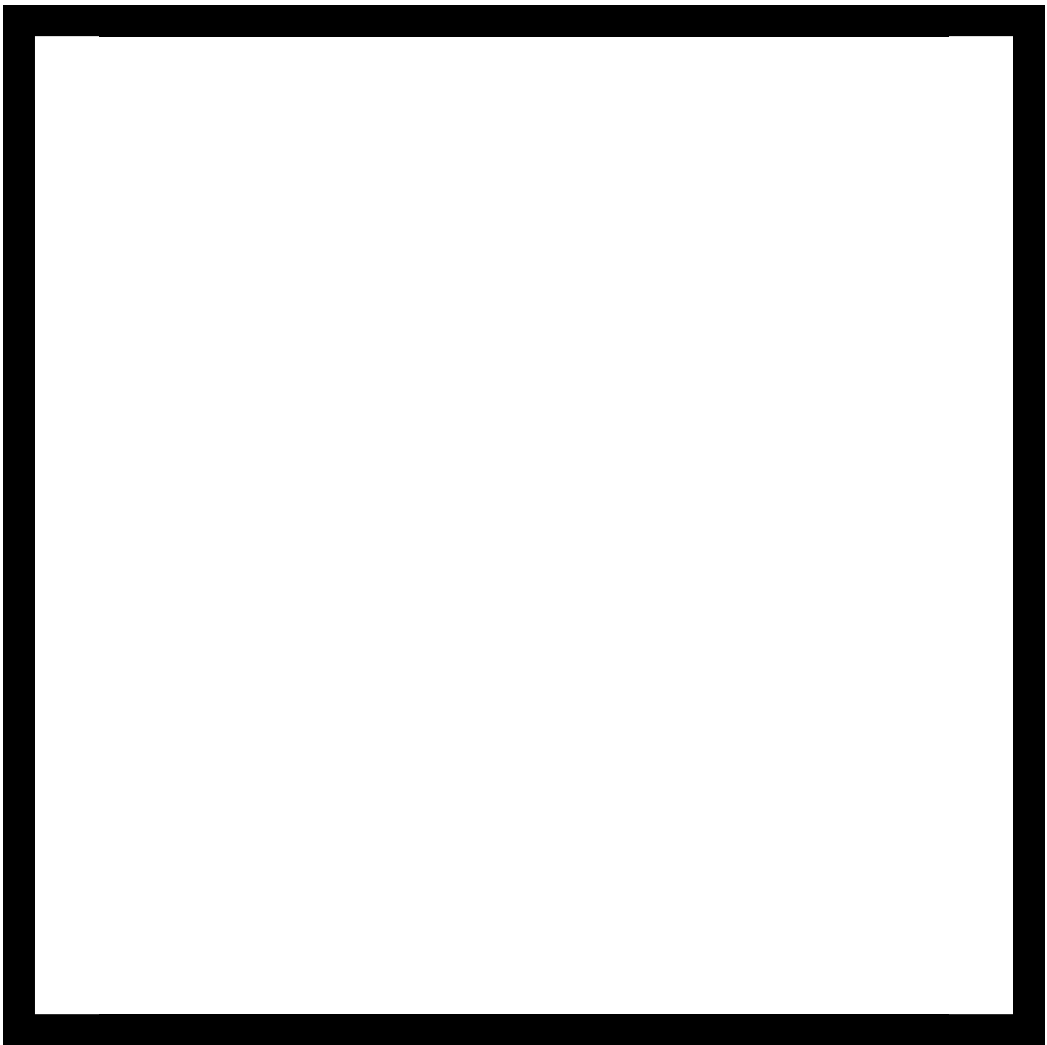


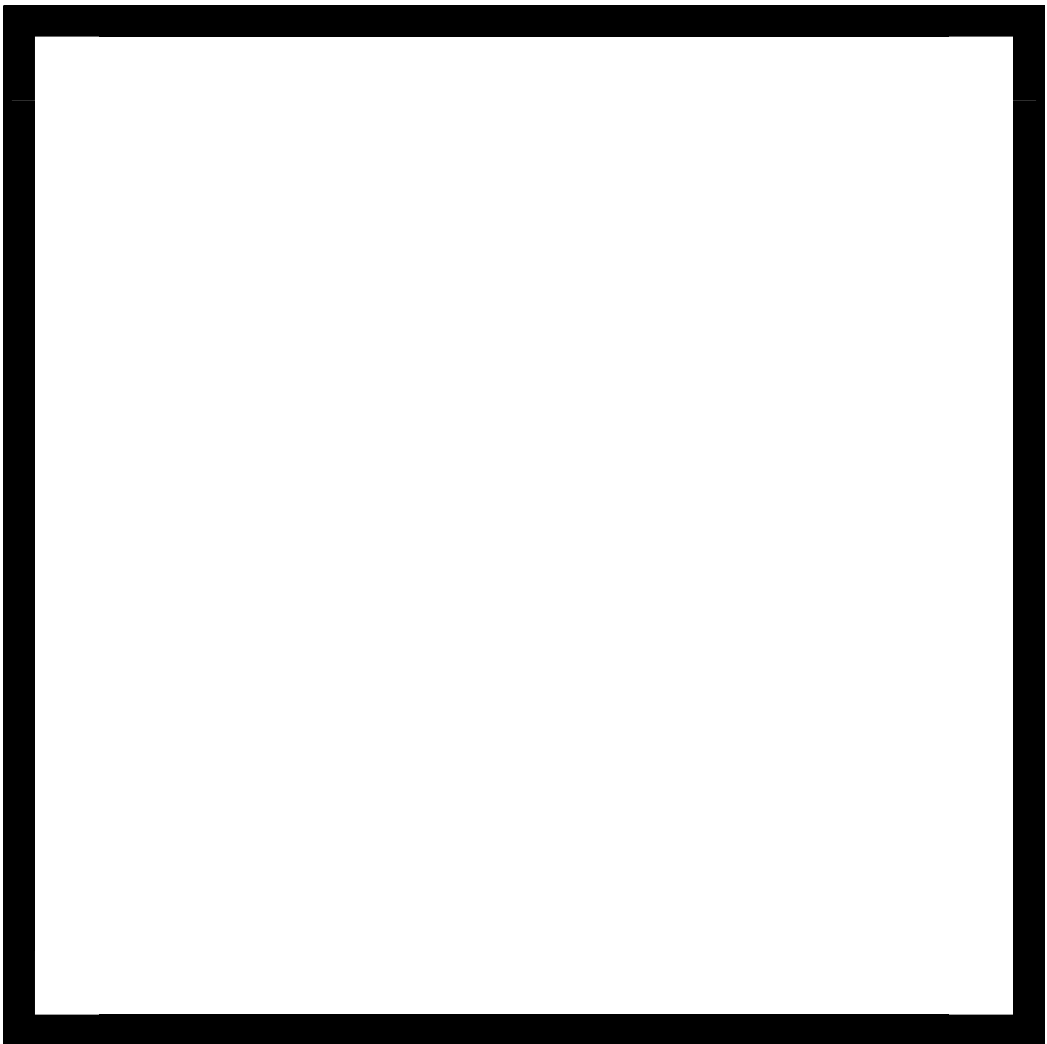
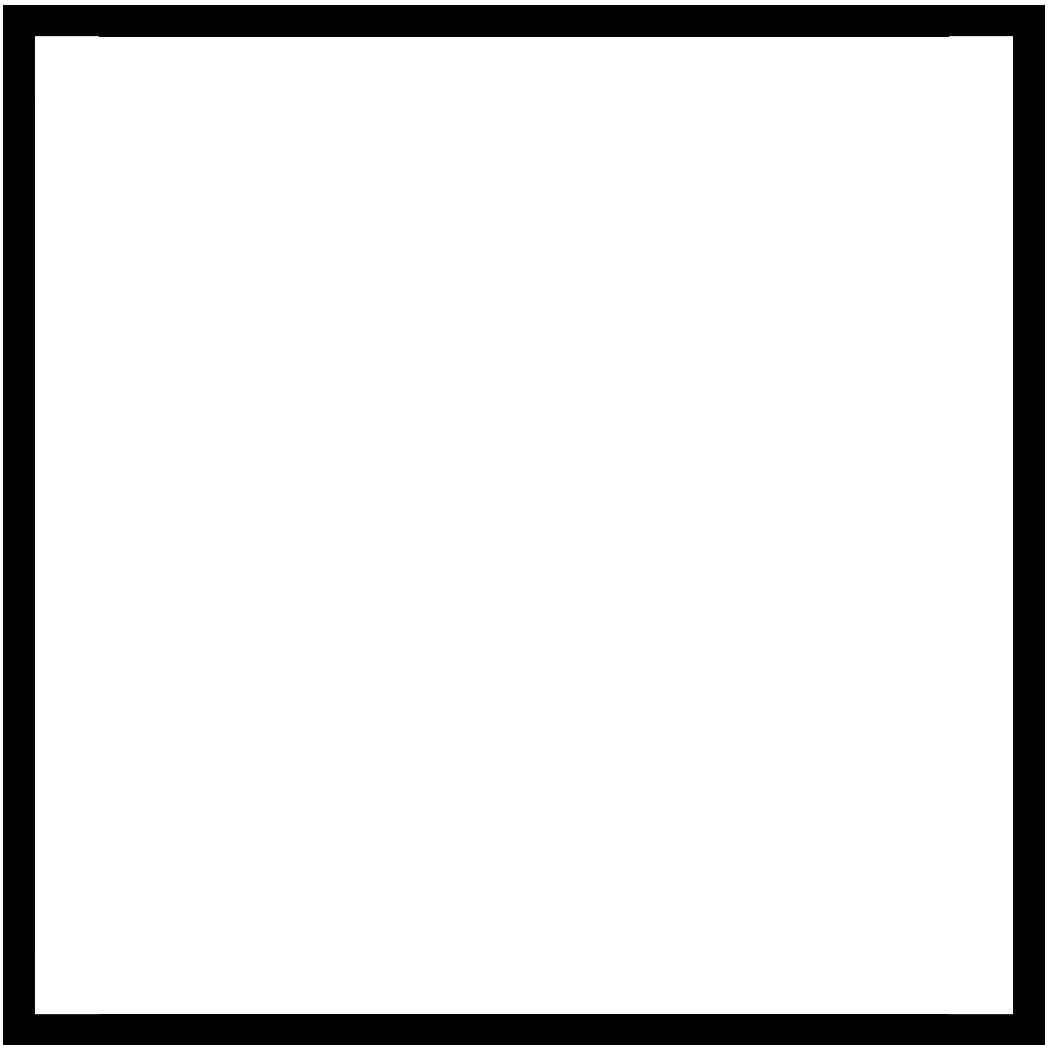








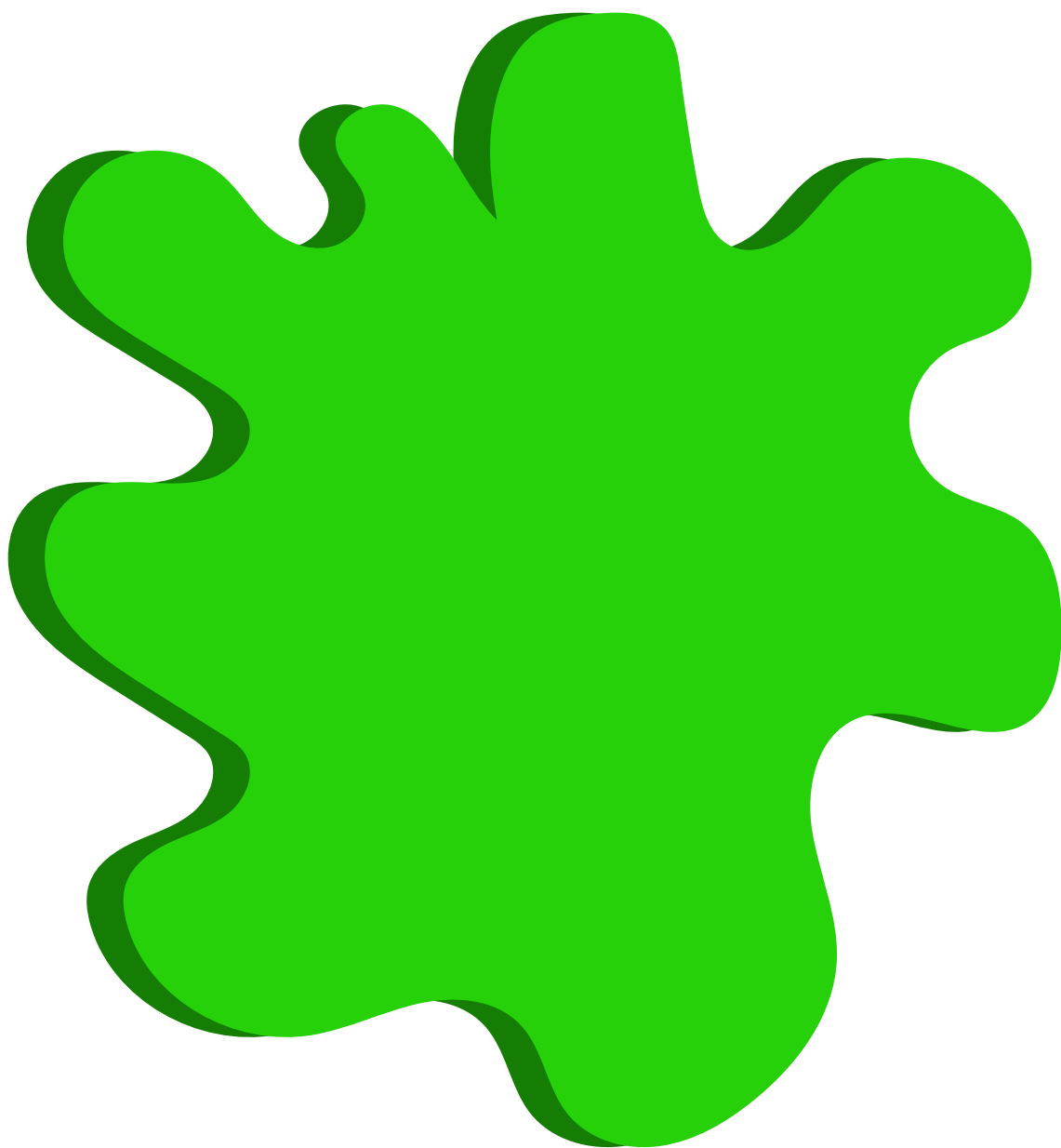






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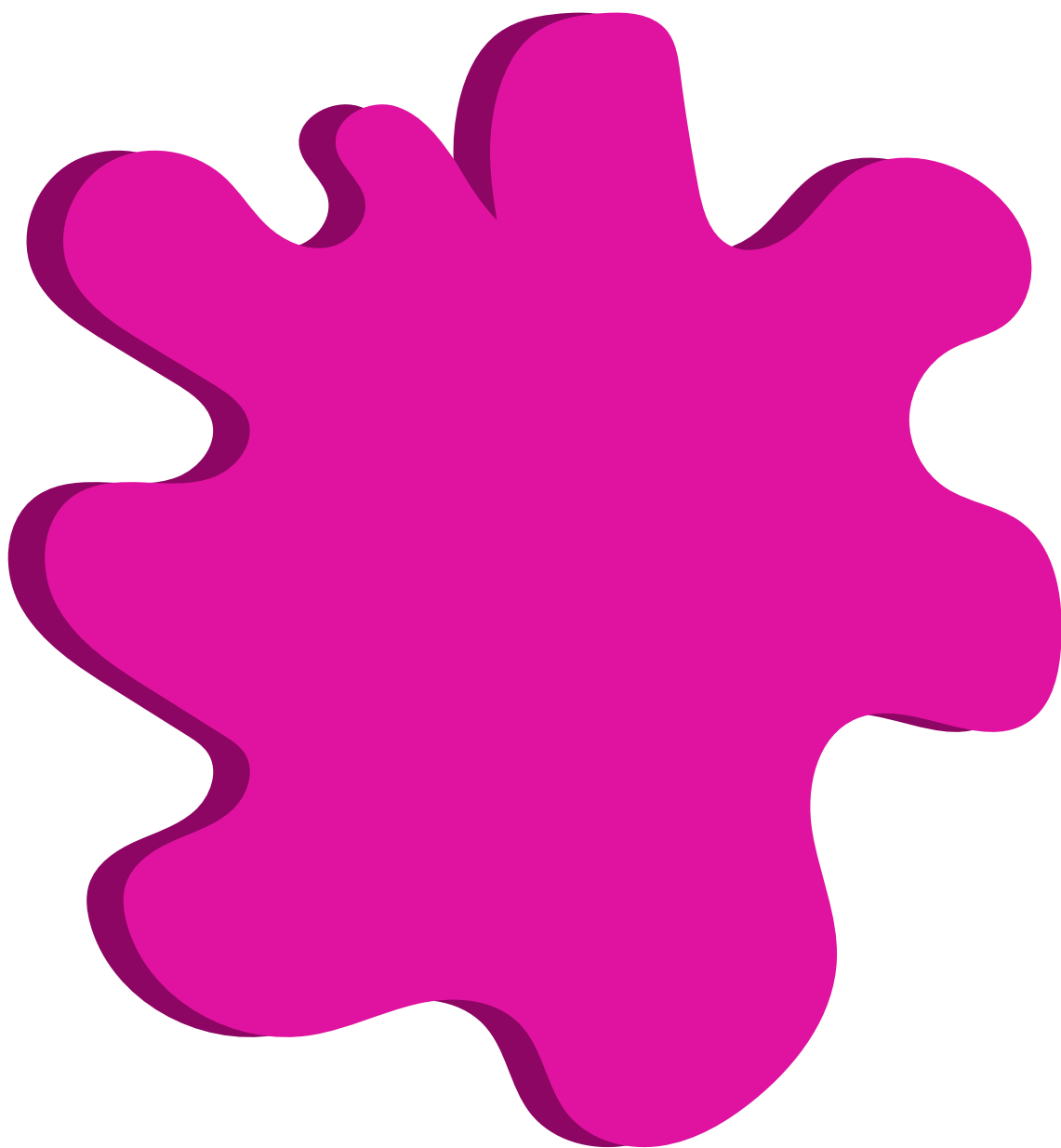






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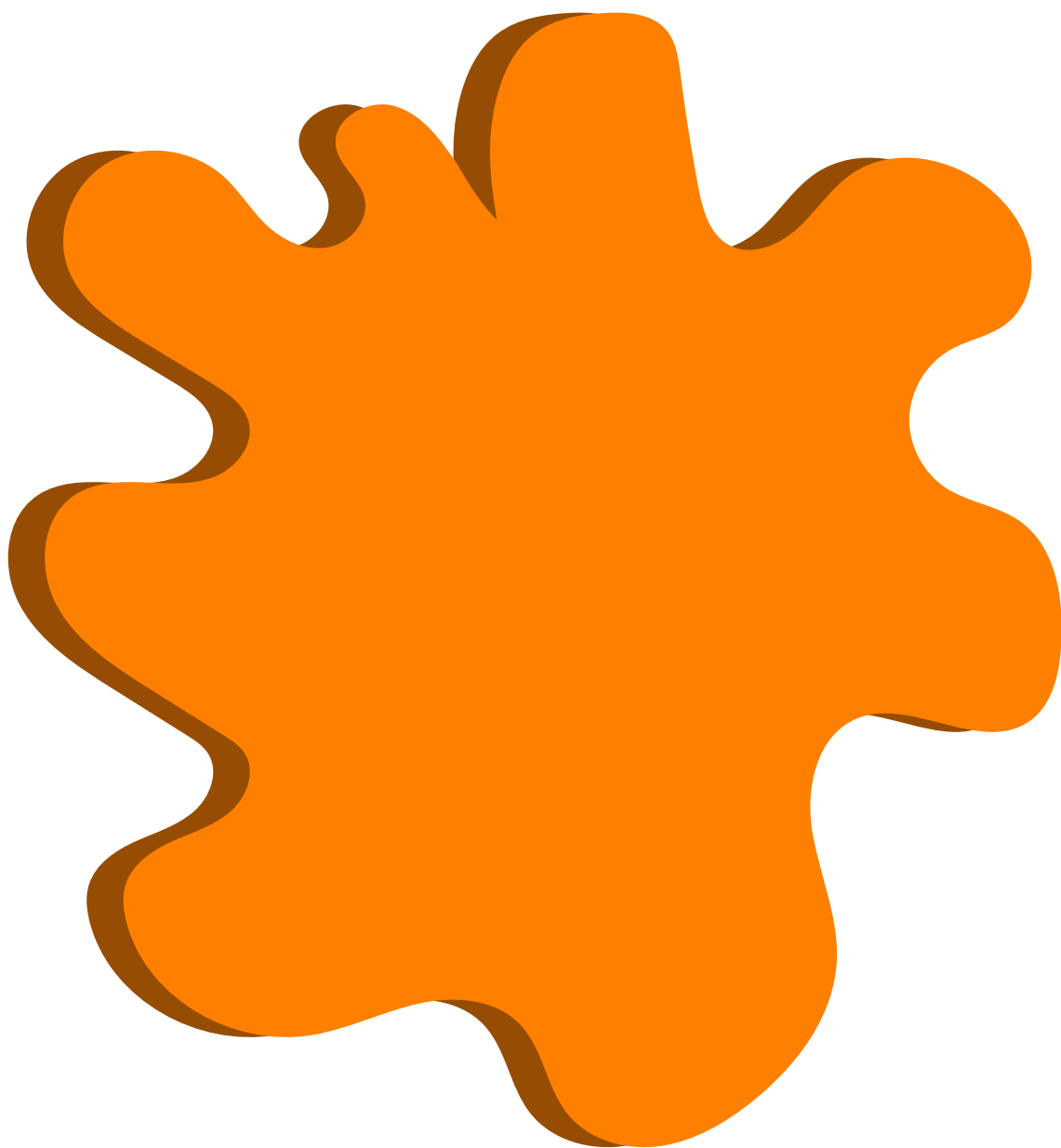






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Mixed colours

